

Cross-Dataset Temporal EEG-Cardiac Coupling Report

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Analyst note: Conservative replication summary based on completed pipeline runs (Stages 0-4). Additional datasets (ds003690, ds003816) and data-driven config tuning are documented below.

Executive summary

We ran the within-subject temporal coupling pipeline on three derivative folders:

Run	Path	Cardiac source	Partitions analyzed
ds003838	derivatives/sum/ds003838	EEG (BIDS)	temporal_rest_coupling memory (n=64)
ds006848	derivatives/sum/ds006848	EEG (BrainVision)	temporal_rest_coupling verbalwmm (n=28)
HIIT-40 PPG	derivatives/sum/hiit40_ppg	PPG	temporal_rest_coupling tetris, post_rest, post_tetris (n=40 each)

In progress (not in replication matrix below):

Run	Path	Cardiac source	Notes
ds003690	derivatives/sum/ds003690	EEG (EEG-AD)	temporal_rest_coupling Requires signal_type: ecg; Stage 2 usable count gated by cardiac QC
ds003816	derivatives/sum/ds003816	EEG (BrainVision)	temporal_rest_coupling Many short tasks; tune min_overlap_s and cardiac windows via --recommend-config-values

Bottom line (conservative read):

- 1. **Coupling strength is detectable descriptively** - median peak $|r|$ is consistently positive across datasets and most partitions (typically 0.05-

0.25), and selected peak-r tests (FDR across 9 pairs) are significant in nearly all partitions. These tests do **not** control for searching over lags.

2. **Permutation-controlled group evidence is absent** - circular-shift null tests (`sig_group_perm_fdr`) did **not** survive FDR in **any** dataset x partition x pair (0/72 partition-pairs checked). Under the stricter evidence layer, we cannot claim group-level coupling beyond what the null would produce after lag search.
3. **Lag direction does not replicate cleanly** - every partition carries a `mixed_peak_lag_direction` warning. Peak-lag vs zero is rarely FDR-significant. Comparing to the **reference anchor** (ds003838 rest), several partitions flip direction (especially HRV-alpha pairs in HIIT post-rest and ds006848 rest).
4. **Task and modality matter** - ds003838 **memory** shows the same broad pattern as rest but at **~3x weaker** peak $|r|$. HIIT **tetris** blocks are weaker than rest blocks. ds006848 (embedded PPG) is weaker and noisier than ds003838 rest (gold-standard ECG).
5. **Stage 4 event-triggered results are usable for HR→EEG** (large event counts), but EEG→HR curves remain exploratory.

Overall replication label: partial / inconclusive - positive descriptive coupling replicates weakly across datasets; timing direction and permutation-supported effects do not replicate strongly enough for causal or directional claims.

Methods snapshot

- **Signals:** z-scored 1 Hz time series of EEG band envelopes (theta, alpha, beta) and cardiac features (HR, RMSSD, SDNN).
- **Stage 3:** Lagged cross-correlation (-60 to +60 s for most runs; ds003838 rest mean curves use -40 to +40 s). Group summaries use median raw peak lag and signed r per subject, then aggregate. Config `peak_selection:` `abs_peak` uses strongest $|r|$ (positive or negative); default `positive_only` ignores negative-only curves.
- **QC warnings:** `mixed_peak_lag_direction` means subjects disagree on whether EEG or cardiac leads - informational, not a failure. High `raw_edge_peaks` suggests widening `lag_max_s` or setting `edge_margin_s`: 5.
- **Inference layers (in order of conservatism):**
 1. Selected peak signed- r vs zero - exploratory (no lag-search correction).
 2. Permutation-controlled group peak strength (circular-shift EEG null) - **preferred** for group claims.
 3. Peak lag vs zero - exploratory; unstable in these cohorts.

- **Stage 4:** Subject-mean-then-group-mean event-triggered averages (HR increase/decrease; theta burst; alpha suppression; beta burst).
- **Convention:** Negative lag → EEG leads cardiac; positive lag → cardiac leads EEG.

Full pipeline documentation: `temporal_coupling/README.md`, `temporal_coupling/IMPLEMENTATION_PLAN.m`

Data coverage and QC

Cohort sizes

Dataset	Partition	Subjects in Stage 3	Observations in audit
ds003838	rest	65	65
ds003838	memory	64	65
ds006848	rest	21	22
ds006848	verbalwm	28	30
HIIT-40	each of 4	40	160 total (20 unique IDs
PPG	partitions		x 8 sessions)

Cardiac modality

- **ds003838:** Dedicated ECG - highest signal quality; used as **reference anchor**.
- **ds006848 & HIIT-40 PPG:** Embedded PPG/photosensor - more artifact-prone; likely contributes to weaker peaks and higher edge-peak rates.
- **ds003690:** Embedded **EKG** (EEGLAB) - label `signal_type: ecg` in config so R-peak detector runs; cardiac QC strongly affects Stage 2 yield.
- **ds003816:** Embedded **ECG** (BrainVision) - large cohort with many **short segments**; alignment yield sensitive to `hr_window_s`, `hrv_window_s`, and `audit.min_overlap_s`.

Data-driven config tuning

After Stage 1b + 1c, run:

```
.venv/bin/python -m ppg_eeg.temporal_coupling \
  --config config.run.<dataset>.temporal_coupling.yaml \
  --recommend-config-values
```

This suggests `min_overlap_s` and cardiac window sizes from `alignment_qc.csv` and overlap-failure patterns. Re-apply to YAML and rerun 1b→1c before interpreting Stage 2 counts (e.g. `wrote=269/375` on ds003690 reflects `usable_for_xcorr`, not total aligned rows).

Edge-peak QC flag

Peaks landing at the search-window boundary suggest unreliable lag estimates.
Pooled edge-peak rates (all pairs):

Dataset	Edge-peak rate (pooled)	Worst pairs
ds003838	3-17%	sdnn__beta (17%), sdnn__alpha (16%)
ds006848	2-12%	sdnn__theta (12%)
HIIT-40 PPG	4-18%	sdnn__alpha (18%), rmssd__alpha (16%)

Stage 3 results by dataset

Reference anchor: ds003838 rest (n=65, ECG)

Strongest and cleanest results in the project so far.

Pair	Median lag (s)	Median peak r	Strength	Direction
hr__alpha	-5	0.236	moderate	EEG leads
hr__beta	+10	0.235	moderate	cardiac leads
hr__theta	0	0.214	moderate	near-zero
rmssd__alpha	-10	0.226	moderate	EEG leads
rmssd__beta	-15	0.255	moderate	EEG leads
sdnn__alpha	-7.5	0.236	moderate	EEG leads
sdnn__beta	-10	0.245	moderate	EEG leads

HRV-EEG pairs show a consistent **EEG-leads-by ~5-20 s** pattern. HR-EEG pairs are moderate with mixed direction. Selected peak-r FDR: all 9/9 significant. Permutation FDR: 0/9. Peak-lag FDR: 0/9.

Plots: derivatives/run_ds003838_temporal_coupling/ds003838/group/rest/group_cross_correlation

ds003838 memory (n=64, ECG)

Same **directional sign** as rest for most pairs, but **much weaker** amplitudes (median r ~ 0.07-0.11 vs 0.19-0.25 at rest). All pairs classified weak-moderate. Only **sdnn__beta** reaches moderate strength.

Interpretation: active memory task **attenuates** coupling magnitude without reversing the broad EEG-leads-HRV pattern. Partial replication of rest **in direction**, not in strength.

ds006848 (embedded PPG)

Rest (n=21)

Pair	Median lag	Median r	vs ds003838 rest
hr__theta	-5 s	0.133	Same EEG-leads direction; ~62% of reference r
hr__alpha	-15 s	0.268	Same direction; similar r (but n=21)
rmssd__alpha	-25 s	0.082	Direction flip (ref: -10 s, EEG leads)
sdnn__alpha	-20 s	0.131	Direction flip (ref: -7.5 s, EEG leads)

Small sample (21 rest subjects with peaks). Several SDNN pairs have high edge-peak rates (19-24%).

Verbal working memory (n=28) Uniformly **weak** coupling (median r ~ 0.04-0.08). EEG-leads direction for most pairs but peak-lag tests non-significant. Descriptive peak-r FDR significant for all pairs; permutation FDR none.

HIIT-40 PPG (photosensor, n=40 per partition)

20 participants x 4 conditions (pre/post x rest/tetris); PS and PH groups pooled per partition.

Rest blocks (pre_rest, post_rest)

- **Moderate** HR-EEG coupling (median r ~ 0.17-0.22) - comparable to, but slightly below, ds003838 rest.
- **pre_rest**: hr__alpha EEG-leads (-5 s, r=0.220) - **matches reference direction**.
- **post_rest**: hr__alpha **cardiac-leads** (+7.5 s, r=0.197) - **opposite to reference**; rmssd/sdnn alpha also flip vs reference in post_rest.
- HRV-theta pairs: EEG-leads in pre_rest; mixed in post_rest.

Tetris blocks (pre_tetris, post_tetris)

- Weaker coupling (median r ~ 0.09-0.15 for HR pairs; many HRV pairs weak).
- ****post_tetris hr__alpha:**** only partition across all datasets with **FDR-significant peak lag** (median +17.5 s, cardiac leads, q=0.011) - but still no permutation support.

- Active gameplay reduces coupling magnitude vs rest segments within the same subjects.

Plots (examples):

- derivatives/run_hiit_40_ppg_temporal_coupling/hiit/group/pre_tetris/group_cross_correla
- derivatives/run_hiit_40_ppg_temporal_coupling/hiit/group/post_rest/group_cross_correlat

Cross-dataset replication matrix

Reference: **ds003838 rest**. Comparison focuses on median peak lag sign and coupling strength.

Comparison	Peak strength	Lag direction	Replication label
ds003838 rest → ds003838 memory	Weaker (~3x)	Mostly same (EEG-leads HRV)	Partial
ds003838 rest → ds006848 rest	Weaker	Mixed (HRV-alpha flips)	Inconclusive (low n, PPG)
ds003838 rest → ds006848 verbalwm	Much weaker	Broadly same sign	Partial (weak)
ds003838 rest → HIIT pre_rest	Similar HR pairs	hr__alpha matches	Partial
ds003838 rest → HIIT post_rest	Similar HR pairs	HRV-alpha flips	Does not replicate (direction)
ds003838 rest → HIIT tetris	Attenuated	Mixed	Inconclusive

Pairs with most cross-dataset consistency

- ****hr__theta:**** EEG-leads (~5 s) in ds003838 memory, ds006848 rest/verbalwm, and all HIIT partitions. Best candidate for a reproducible directional pattern (still without permutation support).
- ****HRV-beta (rmssd__beta, sdn__beta):**** EEG-leads in ds003838 rest and HIIT pre_tetris; inconsistent elsewhere.

Pairs with poorest replication

- ****rmssd__alpha, sdn__alpha:**** Direction flips in HIIT post_rest and ds006848 rest vs reference. High edge-peak rates. Treat lag estimates as

unreliable.

Stage 4: Event-triggered averages

All partitions meet usability thresholds for group plots (≥ 10 total events, ≥ 4 subjects). HR-triggered EEG events are abundant; EEG-triggered cardiac events are sparser in short rest recordings.

Dataset	Partition	HR events/subject	Theta burst/subject
ds003838	rest	~7.5	~1.6
ds003838	memory	~710	~135
ds006848	rest	~58	~10
ds006848	verbalwm	~479	~86
HIIT PPG	rest blocks	~14	~3
HIIT PPG	tetris blocks	~44	~8

Pipeline guidance (consistent across all `stage4_interpretation_notes.txt` files):

- **HR $\rightarrow$ EEG plots:** most reliable Stage 4 output.
- **EEG $\rightarrow$ HR plots:** exploratory only.
- Event curves are z-scored descriptive averages - not causal evidence.

Example plots per partition: `event_triggered_hr_to_eeg.png`, `event_triggered_eeg_to_hr.png` in each `group/{partition}/` folder.

Statistical evidence summary

	ds003838	ds003838	ds006848	ds006848	HIIT (4
Test	rest	memory	rest	verbalwm	partitions)
Selected	9/9	9/9	9/9	9/9	9/9 each
peak-					
r					
FDR					
Peak-	0/9	0/9	0/9	0/9	0/9 (except
lag					post_tetris
FDR					hr_alpha
					1/9)
Permutation	0/9	0/9	0/9	0/9	0/9 each
peak					
FDR					

Conservative conclusion: The only test that controls lag-search bias at the group level does not support coupling in any partition. Selected peak-r significance alone is insufficient for publication-grade claims.

Limitations

1. **Different cardiac sensors** - ECG (ds003838) vs embedded PPG (ds006848, HIIT). Direct amplitude comparison is unfair; direction comparisons are also sensor-dependent.
 2. **Small ds006848 rest n=21** - limits power for group lag tests and permutation inference.
 3. **HIIT post vs pre confound** - exercise intervention may shift autonomic dynamics and coupling direction (seen in post_rest HRV-alpha flip).
 4. **Edge peaks** - 10-24% of peaks at window boundary for some HRV pairs; lag estimates for those pairs should not be interpreted literally.
 5. **Multiple comparisons** - 9 pairs x 8 partitions x 3 datasets; focus on patterns, not individual p-values.
 6. **No causal claims** - all results are descriptive timing within subjects.
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Recommended next steps

1. **Lead with ds003838 rest ECG results** as the internal benchmark; treat PPG datasets as secondary replication attempts.
 2. **Report permutation null results prominently** - do not rely on selected peak-r FDR alone.
 3. **Split HIIT by PS vs PH** (or pre vs post) before drawing intervention conclusions - pooled n=40 may hide condition x direction interactions already visible between pre_rest and post_rest.
 4. **Run ECG-based HIIT partition** (derivatives/run_hiit_40_ecg_temporal_coupling) when available - separates exercise effects from PPG quality effects.
 5. **Automate cross_dataset_temporal_coupling_summary.csv** (planned in IMPLEMENTATION_PLAN) to formalize replication labels per pair. Use --recommend-config-values per dataset before locking window/overlap parameters.
 6. **Stage 4:** prioritize HR-triggered EEG figures for Russell; defer EEG-triggered cardiac figures to supplementary material.
 7. **Complete ds003690 and ds003816** runs with dataset-specific cardiac QC, then add to cross-dataset matrix if effect sizes and directions are stable enough.
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Key file index

Content	Location
ds003838 group summaries	derivatives/run_ds003838_temporal_coupling/ds003838/group/{
ds006848 group summaries	derivatives/run_ds006848_temporal_coupling/ds006848/group/{
HIIT-40 PPG group summaries	derivatives/run_hiit_40_ppg_temporal_coupling/hiit/group/{p
Interpretation notes	stage3_interpretation_notes.txt, stage4_interpretation_notes.txt in each partition folder
Peak statistics	peak_correlation_summary.csv, group_cross_correlation_summary.csv
Mean +/- SEM curves	group_cross_correlation_mean_sem_grid.png
Data audit	derivatives/run_ds003838_temporal_coupling/ds003838/group/d (130 rows) derivatives/run_ds006848_temporal_coupling/ds006848/group/d (52 rows)
Config recommendations	python -m ppg_eeg.temporal_coupling --config <cfg> --recommend-config-values
ds003690 / ds003816 (WIP)	derivatives/run_ds003690_temporal_coupling/, derivatives/run_ds003816_temporal_coupling/

*Report generated from pipeline outputs in the three derivative runs listed above.
Interpretation follows the conservative evidence hierarchy defined in the temporal
coupling module.*